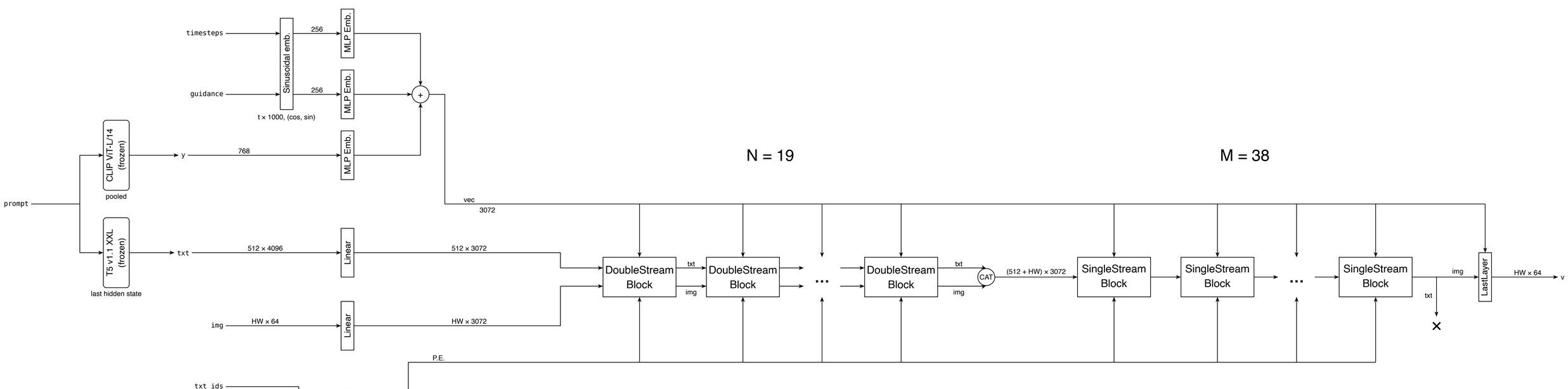
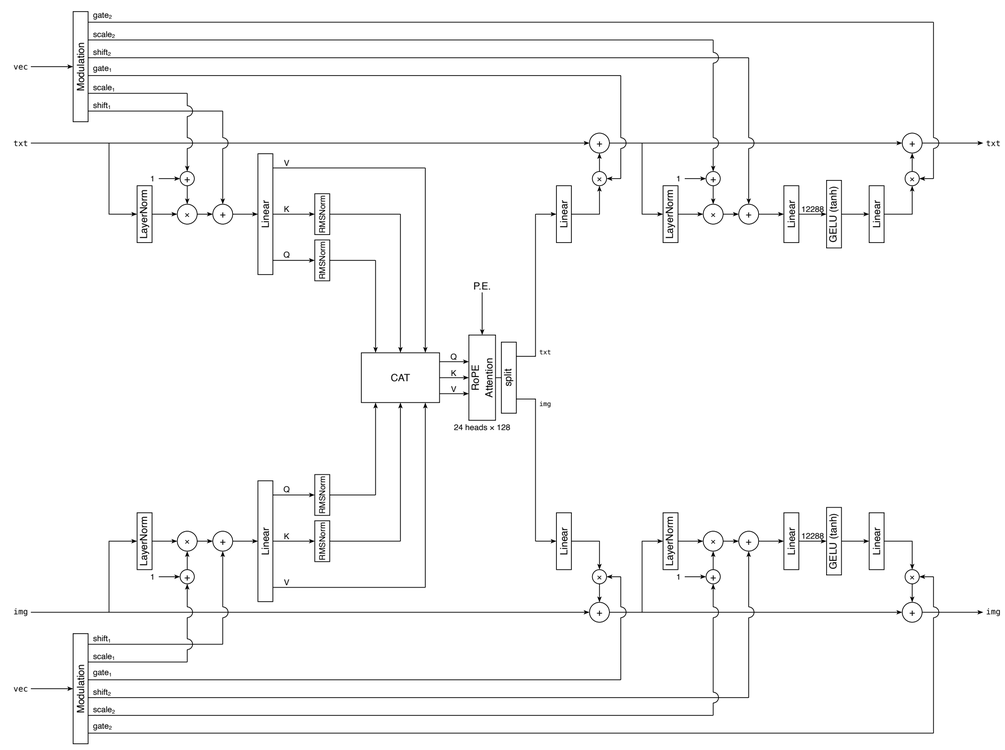
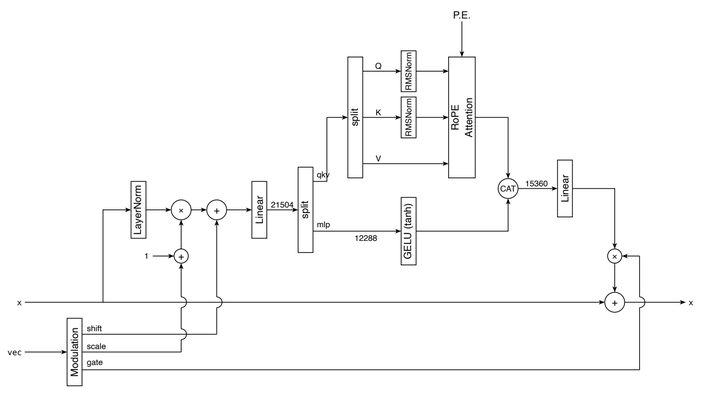


Flux.1 global architecture



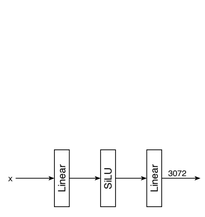
DoubleStream block

separate weights for the txt and img streams, they only meet in the attention

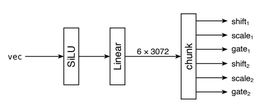


SingleStream block

attention and MLP in parallel out of one Linear

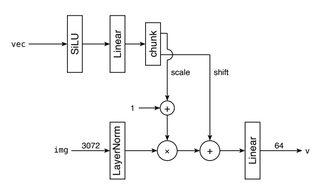


MLP Emb.



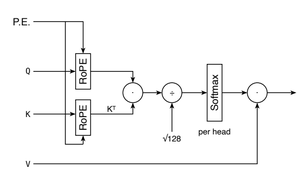
Modulation

two per DoubleStream block (one per stream), one per SingleStream block with only shift, scale and gate (3 × 3072)

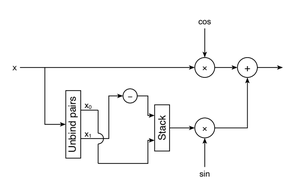


LastLayer

image tokens only

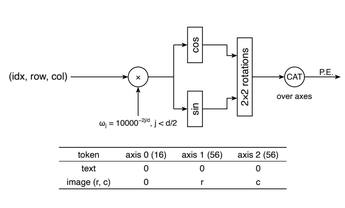


RoPE Attention



RoPE

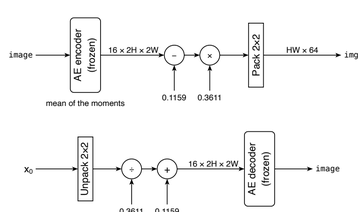
interleaved pairs (2i, 2i + 1), one frequency per pair



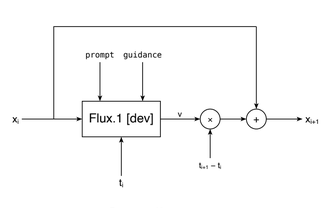
token	axis 0 (16)	axis 1 (56)	axis 2 (56)
text	0	0	0
image (r, c)	0	r	c

EmbedND

axis 0 is always 0 here, Kontext sets it to 1 for the reference image



Latent space



Sampling step

default: 50 Euler steps, guidance 2.5, schedule shifted with the resolution

References

- source code: github.com/black-forest-labs/flux (`model.py`, `modules/layers.py`, `math.py`, `sampling.py`)
- first architecture diagram: @nrehiew on X
- parallel attention and MLP in the SingleStream blocks: Dehghani et al., *Scaling Vision Transformers to 22 Billion Parameters*, 2023
- H, W: latent token grid (image size / 16), the AE latent is 2H × 2W, the T5 prompt is always padded to 512 tokens